

Homework 1

● Graded

Student

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Total Points

36 / 36 pts

Question 1

Problem 1

1 / 1 pt

✓ - 0 pts Correct answer - C

- 1 pt Incorrect answer

Question 2

Problem 2

1 / 1 pt

✓ - 0 pts All options are correct

- 0.5 pts missed two options

- 0.25 pts missed one option

- 0.75 pts missed three options

Question 3

Problem 3

1 / 1 pt

✓ - 0 pts Correct

- 1 pt Incorrect

Question 4

Problem 4

3 / 3 pts

✓ - 0 pts Correct

- 1 pt It should be FALSE

- 1 pt Should define $P(A \cap B) = 0$ for disjoint events A and B

- 1 pt Should use $P(A \cup B) = P(A) + P(B) > 1$ to explain that it is impossible for A and B to be disjoint.

- 3 pts Not seeing the answer

Question 5

Problem 5

3 / 3 pts

✓ - 0 pts Correct

- 1 pt (a) should be False since $P(A^C) + P(B) = 1 - P(A) + P(B) \leq 1$, i.e. $P(B) \leq P(A)$.
- 1 pt (b) should be True since $P(A \cup B) \leq P(S) = 1$
- 1 pt (c) should be True since $P(A \cup C) \leq P(A \cup B \cup C)$
- 0 pts I will take that you meant (b) and (c) are true. Please indicate clearly what your answer is from now on.

Question 6

Problem 6

12 / 12 pts

✓ - 0 pts Correct

- 1 pt (a) answer
- 1 pt (b) answer
- 1 pt (c) answer
- 1 pt (d) answer
- 2 pts (a) Show work / Incorrect reasoning
- 2 pts (b) Show work / Incorrect reasoning
- 2 pts (c) Show work / Incorrect reasoning
- 2 pts (d) Show work / Incorrect reasoning

Question 7

Problem 7

10 / 10 pts

✓ - 0 pts Correct

- 1 pt algebraic error
- 4 pts misunderstood what the probability is supposed to be
- 3 pts incorrect series expression
- 5 pts major errors in problem
- 0.5 pts answer not simplified fully
- 2 pts error in simplifying the series
- 4 pts did not express as a series
- 3 pts did not simplify series expression
- 10 pts not attempted
- 5 pts no work shown

✓ - 0 pts Correct

- 1 pt Should identify that "a person is neither a mathematician nor a coffee lover" is the complement of $P(M \cup C)$

- 1 pt Should use $P(M \cup C) = P(M) + P(C) - P(M \cap C)$

- 1 pt Should get $P(M \cup C) = 0.82$

- 1 pt Should use $P((M \cup C)^c) = 1 - P(M \cup C)$

- 1 pt Should get $P((M \cup C)^c) = 0.18$

1. • Favorite snack, stay home $\Rightarrow 1$

• No snack, stay home and order, or grocery shopping $\Rightarrow 2/3$

• Grocery shopping, take dog, or leave dog $\Rightarrow 3/4$

Final Answer: Option C, or number 2 shows that

Sarah will order online.

1 heads

3/4 tails

tails > heads

2. Option a, b, c, and d are all correct answers.

3. Option b and option d are correct.

$$b) (B - C) \cup (C - B) \cap A$$

$$\begin{array}{ccc} \hookrightarrow \{2, 4\} & \hookrightarrow \{3, 6\} & \rightarrow \{1, 4, 6, 7\} \end{array}$$

only similar elem.

$$\rightarrow \{2, 3, 4, 6\} \rightarrow \text{Intersection with } A: \{4, 6\} \checkmark$$

$$d) (B \cap C' \cap A) \cup (B' \cap C \cap A) \rightarrow \{4, 6\} \checkmark$$

$$P(A \cap B) = 0$$

$$P(A) = 0.7$$

$$P(B) = 0.5$$

1. No, A and B cannot be disjoint because $P(A) + P(B) = 1.2$, which is greater than the maximum possible probability of 1. In order for 2 events to be disjoint, the sum of their probabilities must be less than or equal to 1. ✓

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.7 + 0.5 - 0 = 1.2$$

↳ Probabilities can't be greater than one,

So A and B being disjoint is impossible.

They must overlap, meaning $P(A \cap B) > 0$.

Data 400 Problem set Hwl Con.

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

↑ Pairwise disjoint

5. a) $P(A^c) + P(B) \leq 1$

$$\hookrightarrow 1 - P(A) + P(B) \leq 1$$

$$\hookrightarrow -P(A) + P(B) \leq 0$$

$$\hookrightarrow P(B) \leq P(A)$$

$$A \cap B = B \cap C = A \cap C = \emptyset$$

$$P(A) \leq 1, P(B) \leq 1$$

$$P(A^c) = 1 - P(A)$$

Part a) is not always true, as if $P(A) = 0.2$ and

$P(B) = 0.5$, then $P(A^c) + P(B) = 0.8 + 0.5 = 1.3$, which is greater

than 1, effectively violating the statement.

$\hookrightarrow$ False

↑ No overlap

b) $P(A) + P(B) \leq 1$, disjoint so $P(A \cup B) = P(A) + P(B)$

$$P(A \cup B) = P(A) + P(B) \leq 1$$

Part b) is true because the probability of any event is bound by 1. Adding another event into the sample space doesn't change this as A and B are disjoint and their combined probability can't exceed one.
→ True

c) $P(A \cup B \cup C) \geq P(A \cup C)$, Pairwise disjoint: $P(A \cup B \cup C) = P(A) + P(B) + P(C)$ and
→ $P(A) + P(B) + P(C) \geq P(A) + P(C)$ and $P(A \cup C) = P(A) + P(C)$

This is always true because $P(B) \geq 0$,

so $P(A \cup B \cup C)$ is always at least as big as $P(A \cup C)$.

Even if $P(B) = 0$, the expressions will still be equal

to each other.

→ True

- both means intersection ($\cap$)
- not means complement (exclude)

Data4co problem set Hw1

6. a) $\boxed{0.93}$ $\rightarrow P(A_2) = 0.07$, $1 - P(A_2) = P(A_2^c)$
 $1 - 0.07 = 0.93 \checkmark$

$P(A_2) = 0.07$
 $P(A_3) = 0.05$

b) $\boxed{0.02}$ $\rightarrow P(A_2 \cup A_3) = 0.10$
 $P(A_2 \cup A_3) = P(A_2) + P(A_3) - P(A_2 \cap A_3)$

$0.10 = 0.07 + 0.05 - P(A_2 \cap A_3)$

$P(A_2 \cap A_3) = 0.12 - 0.10$

$P(A_2 \cap A_3) = 0.02 \checkmark$

$P(A_1 \cap A_2 \cap A_3) = 0.01$

c) $\boxed{0.01}$ $\rightarrow P(A_1^c \cap A_2 \cap A_3) = P(A_2 \cap A_3) - P(A_1 \cap A_2 \cap A_3)$
 $\hookrightarrow 0.02 - 0.01 = 0.01$

$\nearrow A_2^c$
 $\nearrow$ left with $-A_1$
 $\nearrow A_2/A_3$ cancel out

d) $\boxed{0.99}$ $\rightarrow P(\text{at most 2}) = 1 - P(A_1 \cap A_2 \cap A_3)$
 $\hookrightarrow 1 - 0.01 = 0.99$

→ Complement of all ≥ 3 defects is having fewer than 3 defects.
 at most 2

Question assigned to the following page: [6](#)

$$7. P(n) = \frac{2}{3^n}, \text{ for } n = 1, 2, 3 \dots$$

$$P(\text{multiple of 5}) =$$

$$E = \{5, 10, 15, \dots\}$$

$$P(E) = \frac{2}{3^5} + \frac{2}{3^{10}} + \frac{2}{3^{15}} + \dots$$

$$\sum_{n=1}^{\infty} \frac{2}{3^n} = \frac{2}{1-1/3}$$

$$= \frac{2}{243}$$

$$\sum_{i=1}^{\infty} \left(\frac{2}{243^i} \right) = 2 \sum_{i=1}^{\infty} \left(\frac{1}{243^i} \right) = \frac{\frac{1}{243}}{1 - \frac{1}{243}} = \frac{1}{242}$$

$$= 2 \sum_{i=1}^{\infty} \frac{1}{242^i} = \frac{2}{1} \cdot \frac{1}{242} = \frac{2}{242} = \frac{1}{121} \checkmark$$

Final answer: $\boxed{\frac{1}{121}}$ $\checkmark$

$$\uparrow P(M \cup C) \quad \uparrow 1 - P(M \cap C)$$



$$8. \quad P(M) = 0.6, \quad P(C) = 0.7, \quad P(M \cap C) = 0.48$$

$$P(M \cup C) = P(M) + P(C) - P(M \cap C)$$

→ removes intersection

$$= 0.6 + 0.7 - 0.48 = 0.82 \rightarrow P(M \cup C) = 0.82$$

$$1 - 0.82 = 0.18 \checkmark$$

$$\boxed{0.18}$$

